STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-REPRESENTATIONS-RS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 525-532 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 534-561, the complete author proof (26 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 28 of queue_examples-defos.json (pb=1041, stmt 525-532, proof 534-561).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0D3J: sources/stacks-examples-defos/pinned-original-source.tex, source lines 483–494. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06J5: sources/stacks-examples-defos/pinned-original-source.tex, source lines 561–561. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DVP: sources/stacks-examples-defos/pinned-original-source.tex, source lines 396–401. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Example 0D3J satisfies the Rim-Schlessinger condition (RS). In particular, $\mathcal{D}ef_{V,\rho_0}$ is a deformation category for any finite dimensional representation $\rho_0 : \Gamma \to GL_k(V)$.*

Proof. Let $A_1 \to A$ and $A_2 \to A$ be morphisms of $\mathcal{C}_\Lambda$. Assume $A_2 \to A$ is surjective. According to Formal Deformation Theory, Lemma 06J5 it suffices to show that the functor $\mathcal{F}(A_1 \times_A A_2) \to \mathcal{F}(A_1) \times_{\mathcal{F}(A)} \mathcal{F}(A_2)$ is an equivalence of categories.

Consider an object

$$((A_1, M_1, \rho_1), (A_2, M_2, \rho_2), (\mathrm{id}_A, \varphi))$$

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of the category $\mathcal{F}(A_1) \times_{\mathcal{F}(A)} \mathcal{F}(A_2)$. Then, as seen in the proof of Lemma 0DVP, we can consider the finite projective $A_1 \times_A A_2$-module $M_1 \times_\varphi M_2$. Since φ is compatible with the given actions we obtain

$$\rho_1 \times \rho_2 : \Gamma \longrightarrow \mathrm{GL}_{A_1 \times_A A_2}(M_1 \times_\varphi M_2)$$

Then $(M_1 \times_\varphi M_2, \rho_1 \times \rho_2)$ is an object of $\mathcal{F}(A_1 \times_A A_2)$. This construction determines a quasi-inverse to our functor. $\square$